Sanjana Shah

Ahmedabad, India | Open to relocation | Immediate Joiner
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SUMMARY

Software engineer with 2+ YOE at Meta and three summer internships at Amazon, building AWS backend services, mobile networking infrastructure, and full-stack tools using C++, Java, and Python. Seeking backend and infra engineering roles in India.

EXPERIENCE

Meta Mar 2024 – Aug 2026
Software Engineer New York, NY, USA

- Cut global load-balancing convergence time by 15 minutes and reduced PoP CPU overprovisioning by 5%, saving an estimated $2–3M annually, by building and rolling out a C++ DNS cache-refresh timer for Instagram and Threads on iOS and Android, validating disaster-recovery behavior through Python telemetry and A/B testing.
- Reduced compressed binary size by 438–472 KiB per app across three production iOS apps by analyzing dependencies with Buck, validating dead code through breakpoints and diagnostic logging, and removing the Liger library dependency.
- Sped up image and video requests by 6% across Instagram, Messenger, Threads, and WhatsApp by rolling out mbedTLS ARMv8 AES acceleration on iOS and Android, validated through microbenchmarks, APK testing, and regression monitoring.
- Unified network security across 15+ Meta apps by designing, A/B testing, and deploying a centralized HTTP request header security feature spanning Android (Java, Kotlin), iOS (Objective-C), and native (C++) stacks.
- Improved debugging of network and user-experience issues, including 15+ SEV investigations, by building networking bug-report extensions with diagnostic logging, and owning their rollout across Messenger, Threads, and WhatsApp, with A/B testing for memory regressions.

Amazon May – Aug 2021, 2022, 2023
Software Development Engineering Intern Seattle, WA, USA

- Cut per-ticket handling time by 5 minutes per developer and sped up infrastructure ticket access for 4 customer teams by building an automated ticketing system for fault localization and routing on top of my prior-summer monitoring service, using Step Functions, Lambda, and EventBridge Pipes and scoping requirements with 8 stakeholders through prototypes.
- Enabled system-health analysis across 6 team services by building the underlying monitoring service, which categorizes ML infrastructure logs through event-driven Step Functions and EventBridge workflows.
- Let additional services integrate with the monitoring service by designing its infrastructure to be modular using SNS and DynamoDB Streams.

EDUCATION

Arizona State University Jan 2023 – Dec 2023
M.S. Computer Science (Thesis) | GPA: 4.0

- Thesis: Modeling Spatial Competition and Therapy Protocols in Three-Dimensional Vascularized Cancers (C++ agent-based simulator).

Arizona State University Aug 2019 – Dec 2022
B.S. Computer Science | Summa Cum Laude, GPA: 4.0

- Achievements: Moeur Award for 4.0 GPA and New American University Scholar.
- President, Women in Computer Science: revived a 100+ member chapter, led hackathons, speaker events, and mentorship programs.

SKILLS

Languages & Frameworks: C, C++, Java, Kotlin, Objective-C, Python, React, TypeScript, JavaScript (D3.js), HTML, CSS, SQL.

Backend & Infrastructure: AWS Lambda, Step Functions, EventBridge, SNS, DynamoDB, REST APIs, Spring Boot, PostgreSQL, distributed systems, CI/CD.

Systems & Reliability: TLS, DNS, transport protocols, mbedTLS, mobile networking, performance profiling, telemetry, A/B testing.

Developer Tools: Git, GitHub, Linux/Unix, Claude, Codex, GitHub Copilot.

PROJECTS

DEVS-Suite Log Visualizer Java, Spring Boot, PostgreSQL, JavaScript, D3.js

Enabled analysis of timestamped simulation logs through time-based and piecewise-constant charts by building a full-stack application with Spring Boot, PostgreSQL, and D3.js.